

RESEARCH ARTICLE

Seasonal Exposure to Hemispheric Conservation Challenges Influences Population Trends of Migratory Warblers (Parulidae)

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ABSTRACT

Human activities have profound impacts on ecological systems and biodiversity worldwide. Estimating wildlife response to those activities across broad spatial scales is challenging, yet effective conservation measures require an understanding of where and when these activities are contributing to population declines. We investigated the relationship between exposure to human activities (conservation challenges) and population trends of migratory warblers (Parulidae) that breed in the United States and Canada. Our four-stage approach (1) summarizes the literature describing the sensitivity of species to conservation challenges; (2) estimates weekly species' exposure across the full annual cycle; (3) quantifies spatial correlations among species' exposure to conservation challenges; and (4) quantifies the relationships between seasonal exposure to conservation challenges and warbler population trends. Exposure during post-breeding migration, followed by the stationary non-breeding season and pre-breeding migration, explained the most variation in warbler population trends. Within the post-breeding migratory season, the conservation challenges with the greatest total seasonal exposure values were roads, light pollution, communication towers, forest management, and livestock management. During this season, species undergoing the steepest declines were associated with exposure to light pollution, communication towers, urban and suburban areas, livestock management, and agriculture. Notably, exposure to conservation challenges during the breeding season was not an important predictor of warbler population trends. Our results provide hemispheric, full annual cycle information for prioritizing conservation investments and additional research for warblers, specifically the importance of addressing challenges to which birds are exposed during post-breeding migration. More broadly, this framework can be used to assess spatiotemporal conservation challenges impacting migratory species trends globally.

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1 | Introduction

Human activities have profound consequences for life on Earth, including the degradation of the biosphere's biological and functional integrity (Richardson et al. 2023). One striking manifestation of this ecosystem degradation over the last two centuries has been the decline of many migratory species across the globe (Wilcove and Wikelski 2008). The United Nations Convention on the Conservation of Migratory Species of Wild Animals (CMS) was established in 1979 to address the international nature of these declines (Baldwin 2011). Despite numerous international agreements through CMS, a recent report shows that migratory species continue to decline, and over 20% face extinction (UNEP-WCMC 2024). These findings underscore the need to improve our understanding of which human activities are contributing to population declines, where and when they have the biggest impact, and how best to manage such processes at relevant spatiotemporal scales. By addressing these knowledge gaps, conservation measures are more likely to be effective at reversing population declines and facilitating the recovery of migratory species globally (Runge et al. 2014).

The plight of migratory species is exemplified by migratory birds, many of which are declining more rapidly than their non-migratory counterparts (Bairlein 2016; Koleček et al. 2021; Rosenberg et al. 2019; Vickery et al. 2014). Habitat degradation, habitat conversion, and climate change are among the major stressors causing precipitous declines in migratory bird numbers across North America and Europe (Bairlein 2016; Rosenberg et al. 2019); these stressors are the result of human activities that are often referred to as threats (International Union for Conservation of Nature (IUCN) 2022), albeit many of the same activities (e.g., agriculture and urbanization) can also offer solutions to the biodiversity crisis if implemented in ways that improve the availability of safe habitat (e.g., bird-friendly agriculture; Chappell and LaValle 2011; Knapp et al. 2021). To acknowledge this duality, we refer to human activities with potentially positive or negative effects on biodiversity as conservation challenges. Conservation challenges include both sources of direct mortality, such as collisions with buildings in urban areas or cars on roadways (Loss, Will, Loss, et al. 2014; Loss, Will, and Marra 2014), and sources of indirect effects, such as the effects of drought or agricultural pesticides on reproductive success, and both are critical to understanding the underlying drivers of biodiversity loss (Nemes et al. 2023).

North America alone has lost an estimated 2.5 billion migratory birds since 1970 (Rosenberg et al. 2019). These birds travel long distances (hundreds to thousands of kilometers) annually between breeding and stationary non-breeding grounds, exposing them to myriad challenges. To address widespread declines in migratory birds, the effect that conservation challenges have on migratory bird population trends must be understood across the full annual cycle, as processes in one location and season can have consequences in distant locations during other times of the year (i.e., carryover effects; Small-Lorenz et al. 2013; Marra et al. 2015; Buchan et al. 2021). Identifying the conservation challenges that are associated with population trends of migratory birds is further

complicated because many of these challenges are spatially and temporally correlated, and the resulting impacts are often cumulative (Geary et al. 2019; Given and Norton 1993; Nemes et al. 2023; Norris et al. 2004).

The availability of global datasets creates new opportunities for understanding which conservation challenges are linked to population trends. Advances in remote sensing technologies have enabled changes in human activities to be rapidly mapped and quantified across the globe, enabling new inferences on where and when species are exposed to them (Buchan et al. 2022). Similarly, massive participatory science programs, such as eBird and iNaturalist, produce occurrence datasets that can be used to map global species distributions (Feldman et al. 2021; Fink et al. 2023). In particular, eBird is now producing distribution models with global coverage at weekly time intervals (<http://ebird.org/science/status-and-trends>, Fink et al. 2023). These temporally explicit distribution maps provide a finer assessment of the co-occurrence of conservation challenges and migratory birds throughout the annual cycle than was previously possible.

North American warblers (defined as those that breed in the United States and Canada; Parulidae) are a charismatic, diverse, and declining species group with a broad variety of geographic distributions, migration distances, habitat associations, and nesting behaviors (Terrill et al. 2020); they also exhibit a wide range of population trajectories (Saracco et al. 2008). The most steeply declining species are already the focus of full annual cycle studies (Buehler et al. 2007; Kramer et al. 2018). However, despite numerous efforts to quantify the effects of conservation challenges at local scales, these have not led to changing population trends (Rosenberg et al. 2019).

Here, we investigate the relationships between the exposure to conservation challenges across the full annual cycle and population trends of North American warblers. To do this, we developed a four-stage approach to: (1) summarize evidence from the literature describing the sensitivity of species to conservation challenges; (2) combine hemispheric maps of conservation challenges derived from remote sensing data with eBird abundance maps to estimate weekly species' exposure across the full annual cycle; (3) quantify spatial correlations among species' exposure to conservation challenges using a principal component analysis; and (4) quantify the relationship between seasonal exposure and conservation challenges (summarized by principal components) and population trends. Our results provide hemispheric, full annual cycle information for prioritizing conservation investments and additional research for warblers.

2 | Materials and Methods

2.1 | Species Selection

For our analysis, we considered 48 warbler species that breed in the United States and Canada. We removed five species due to the lack of available trend estimates from the North American Breeding Bird Survey (BBS) Dataset (Sauer et al. 2020), resulting in 43 total species (hereafter warblers, Table S1). We extracted

survey-wide trend estimates (1966–2022) from the BBS dataset (hereafter, population trend) for each species and classified species into four trend categories to enhance the interpretation of how conservation challenges influence population trends: declining ($< -0.5\%$ per year), slightly declining ($-0.5\% - 0\%$ per year), slightly increasing ($0\% - 0.5\%$ per year), and increasing ($> 0.5\%$ per year).

2.2 | Scoring Sensitivity to Threats

Our work is based upon the conservation lexicon developed by Salafsky et al. (2008) and subsequently used in multiple conservation planning frameworks. The Direct Threats Classification (Conservation Measures Partnership 2016) provides a standardized hierarchical list of threats at level 1 (primary environmental drivers) and level 2 (sub-categories of primary drivers) with a recommendation to add refined level 3 threat categories specific to one's own conservation targets. In this framework, threats are defined as human activities that negatively impact conservation targets (Salafsky et al. 2008). Sensitivity (also called severity in the Direct Threats classification framework) is the extent to which stresses linked to threat exposure negatively impact a species. Based on a combination of literature review and expert knowledge within this framework, we developed a list of 88 threats relevant to migratory birds (Table S2). Stresses are the attributes of a conservation target's ecology that are impaired by threats (Salafsky et al. 2008). We generated a list of seven stresses adapted from the list used by IUCN (International Union for Conservation of Nature (IUCN) 2012) (Table S3). Next, to identify a subset of threats to which each species was sensitive to, we conducted a literature review and recorded when a threat was described as having a negative impact on a species and the stress that it caused (hereafter threat sensitivity matrix, Data S4-S5). In cases where a threat sensitivity was scored for a given species, but there was no information for a similar related species (e.g., same genus or similar life history characteristics), we extrapolated sensitivity to the similar related species that lacked information. We included a species in our analysis if it had a recorded negative impact from a threat from any stress (a sensitivity score of 1). Our full methodology is described in the Supporting Information (Appendix S1).

2.3 | Summarizing Threats Into Conservation Challenges

We pooled threats into 19 hemispherically mappable conservation challenges (e.g., agricultural pesticide/herbicide application and agricultural development threats were merged into the conservation challenge of agriculture). We reduced this list to 17 conservation challenges by removing those predicting future changes (sea level rise, increasingly frequent flooding), then further reduced this set to 12 conservation challenges by excluding those for which $< 10\%$ of warblers were sensitive (power lines, groundwater depletion, water quality, coastal modification, coastal disturbance; Table 1).

2.4 | Mapping Conservation Challenges

We mapped each conservation challenge by identifying spatial layers with hemispheric coverage that were recently published

(the majority within the last 10 years; Table 1), had suitable spatial resolution (3 km or finer when possible) and best represented human activity in a spatial format. We aggregated these layers into 3-km cells aligned to match the spatial resolution, projection, and origin of eBird abundance models, then thresholded each layer (see Appendix S2 for threshold details) into a binary presence/absence surface to represent the footprint of the conservation challenge across the hemisphere.

We visually assessed the accuracy of the hemispheric conservation challenge layers for alignment with national patterns by comparing available datasets and using local expert knowledge. This assessment assumed that national datasets typically offer greater spatial and thematic resolution than global spatial products (Liu et al. 2022; Tulbure et al. 2022). We compared layers in the United States and Canada to other available data for these countries (Table S4) and spot-checked the layers where possible using local expert knowledge. For Latin America, we visually compared the conservation challenge maps with recent official geographic information for Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Ecuador, and Mexico (Table S4). We used hemispheric conservation challenge layers that best matched the spatial patterns in national datasets. As necessary, we merged alternative sources for specific countries to minimize observed discrepancies (Table 1; Appendix S2).

2.5 | Summarizing Exposure

We measured exposure as the weekly percentage of a species' abundance that spatially overlaps (i.e., co-occurs) with a conservation challenge, then summarized exposure into 50-km hexagonal bins. We calculated by hexagon: (1) weekly total abundance of each species, and (2) weekly exposed abundance of each species, yielding the weekly percent of birds exposed by hexagon for each species-conservation challenge combination. Next, we summed the hexagonal data to derive the weekly percent of species' abundance exposed across their ranges, and then plotted the weekly exposure values for each species (Figure 1). We calculated seasonal conservation challenge exposure by grouping weekly warbler exposure values into stationary non-breeding, pre-breeding migration, breeding, and post-breeding migration using pre-defined season dates from eBird weekly abundance models (Fink et al. 2023), then summing exposure across weeks in the season. The resulting values accounted for the variation and duration of exposure over the season.

We retained exposure values for those conservation challenges with a documented negative impact in the threat sensitivity matrix (a sensitivity score of 1; Appendix S1); for challenges without a documented negative impact (scores of 0 and 0.5), we set the exposure for that species $\times$ conservation challenge combination to zero for all weeks. Although we considered conservation challenge sensitivity to be relevant to warblers throughout the annual cycle, there were three challenges related to collisions (light pollution, communication towers, and wind turbines) that were relevant only during the migratory periods (Cabrera-Cruz et al. 2018; Longcore et al. 2013); for these, we set exposure to zero during the breeding and stationary non-breeding periods.

TABLE 1 | The 12 conservation challenges that were mapped hemispherically and the related IUCN levels 1, 2, and 3 threats for which warbler sensitivity was scored.

Conservation challenge	Linked level 1, 2, and 3 threats	Data sources
Urban areas	L1: residential and commercial development L2: housing and urban areas L3: high-density urban development, urban/suburban pesticide/herbicide application, industrial development, airport development, domestic cats	(Microsoft, 2019/ 2024 , 2018/ 2024 ; Sirko et al. 2021 ; WorldPop 2020)
Suburban areas	L1: residential and commercial development L2: housing and urban areas L3: low-density urban development, urban/suburban pesticide/herbicide application, domestic cats	(Microsoft, 2019/ 2024 , 2018/ 2024 ; Sirko et al. 2021 ; WorldPop 2020)
Agriculture	L2: annual and perennial non-timber crops L3: agriculture pesticide/herbicide application, agricultural mowing and harvesting, agriculture development	(Karra et al. 2021)
Livestock management	L2: livestock farming and ranching L3: grazing of natural vegetation, pasture development	(Ramankutty et al. 2008 , 2010 ; Robinson et al. 2014)
Oil and gas production	L2: oil and gas drilling L3: onshore oil and gas development, onshore oil and gas production	(United States Department of Homeland Security 2019)
Wind turbines	L3: onshore wind turbine installations	(Dunnett et al. 2020 ; Hoen et al. 2018)
Roads	L2: roads and railroads L3: roadways, noise pollution	(Meijer et al. 2018)
Communication towers	L3: communication towers	(Unwired Labs 2023)
Forest management	L2: wood and pulp plantations L3: non-native tree plantations, native tree plantations, logging and wood harvesting, inadequate forest management	(Curtis et al. 2018)
Surface water management	L2: dams and water management/use L3: dams, overuse of surface water, changes to natural hydrologic flow	(Conservation Biology Institute 2010 ; Grill et al. 2019 ; Vörösmarty et al. 2010)
Light pollution	L3: light pollution	(Earth Observation Group, Payne Institute for Public Policy 2023 ; Elvidge et al. 2013 , 2021)
Drought	L3: increasingly severe/frequent droughts	(Beguería et al. 2010 , 2014 ; Vicente-Serrano et al. 2010)

2.6 | Relating Population Trends to Seasonal Exposures to Conservation Challenges

To better understand the potential population-level impacts of the 12 conservation challenges, we quantified the relationships between seasonal exposure to conservation challenges and warbler population trends. We first considered the correlation structure among seasonal conservation challenge exposure by examining correlation plots. We then used principal components analysis (PCA) to summarize the correlated conservation challenges into a smaller number of variables. We conducted a separate PCA for the conservation exposure

values for each season. We used a Monte Carlo test with 1000 permutations to select principal components with observed eigenvalues that were significantly ($p < 0.05$) different from a random distribution of eigenvalues, and we report the amount of variation in exposure to conservation challenges that those components explain (McGarigal et al. [2013](#)). To further understand and interpret principal components, we summarized PCA loadings, representing the correlation coefficients between the principal components and exposure to conservation challenges for each season. For each season, we plotted warblers ($n = 43$) on the axes of the first two principal components (biplots) to better understand the relationship among warbler

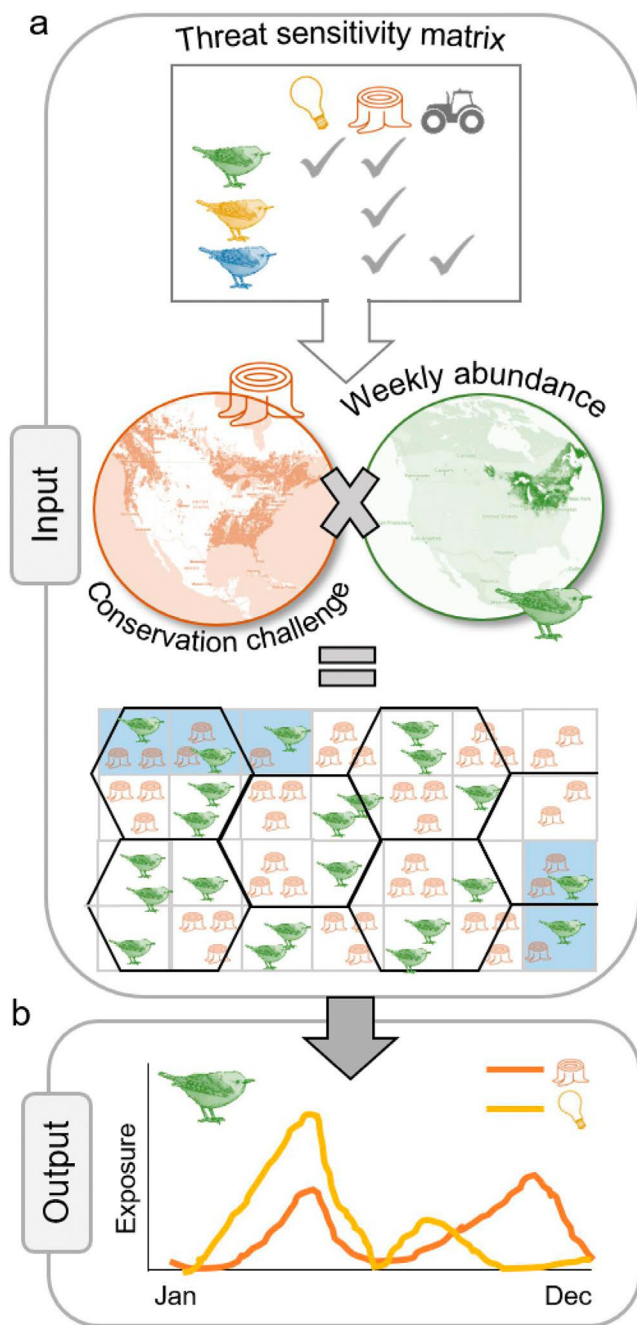


FIGURE 1 | Major input data and spatial calculations, leading to estimates of exposure throughout the annual cycle for North American warblers. (a) Threat sensitivity matrix surfaces the relevant conservation challenges for each species, and those conservation challenge spatial layers are intersected with species' weekly abundance layers at the pixel scale. In pixels where species and conservation challenges co-occur, exposed abundance of birds is compared to total abundance to estimate the percent exposed in each hexagon. (b) This spatial analysis produces range-wide weekly exposure estimates for each species $\times$ conservation challenge combination, which can be further assessed to inform targeted conservation actions during potentially high mortality periods of the life cycle.

species and specific conservation challenges. We then used the principal component scores as a parsimonious set of independent variables that represented warbler exposure to conservation challenges.

We used the retained principal components as independent variables in generalized additive models (GAMs) performed separately for each season to assess the extent to which seasonal exposure to conservation challenges explained variation in warbler population trends previously described above for species selection. GAMs were fit with a Gaussian distribution in R version 4.3.3 using the "gam" function with the "GCV.Cp" smoothing method and $k=9$ in the "mgcv" package (R Core Team 2024). We built separate models per season for two reasons. First, exposure to conservation challenges changes seasonally. It is common for the migration tracks of species to be very different between pre- and post-breeding migration. Therefore, even though the spatial footprints of the conservation challenges do not change seasonally, the distribution for a species is commonly very different between the two migratory periods, leading to potential variation in the strength of the relationship between conservation challenges and trends among seasons. Second, the relationship (both direction and magnitude) between the challenge and population may also change seasonally (i.e., non-stationarity of relationships). For example, livestock may be managed to improve a warbler population during one phase of their annual cycle and could be managed differently during another phase that could lead to population declines. Thus, season-specific results are better suited for conservation and management purposes. We compared seasonal models to determine the relative importance of seasonal exposure to conservation challenges in explaining variation in warbler population trends. Because conservation challenges were explicitly selected for inclusion in the analysis based on a thorough literature review and subsequently reduced to a parsimonious set of exposure variables with the PCA, we considered each season's model as the best representation of the relevant conservation challenges. We compared the deviance explained for each season's GAM to determine which season's exposure to relevant conservation challenges had the strongest relationship with warbler population trends. Within each season, we used hierarchical partitioning to report the percent of deviance explained by each predictor variable using the "gam.hp" package in R (Lai 2024). Finally, we used the GAM partial regression plots to interpret the seasonal relationships among warblers, conservation challenges, and population trends.

3 | Results

3.1 | Sensitivity and Exposure to Conservation Challenges

The threat sensitivity database records (Data S4) were grouped into 12 conservation challenges for 43 species (threat sensitivity matrix; Data S5). Surface water management was listed as a sensitivity for the fewest species (8), while all species (43) were listed as sensitive to urban areas, suburban areas, communication towers, wind turbines, and light pollution. The median number of conservation challenge sensitivities per species was 10.

Exposure varied across warbler species and the annual cycle (Figure 2, Figure S1). Some species, such as the Kentucky Warbler (*Geothlypis formosa*) were most exposed to challenges

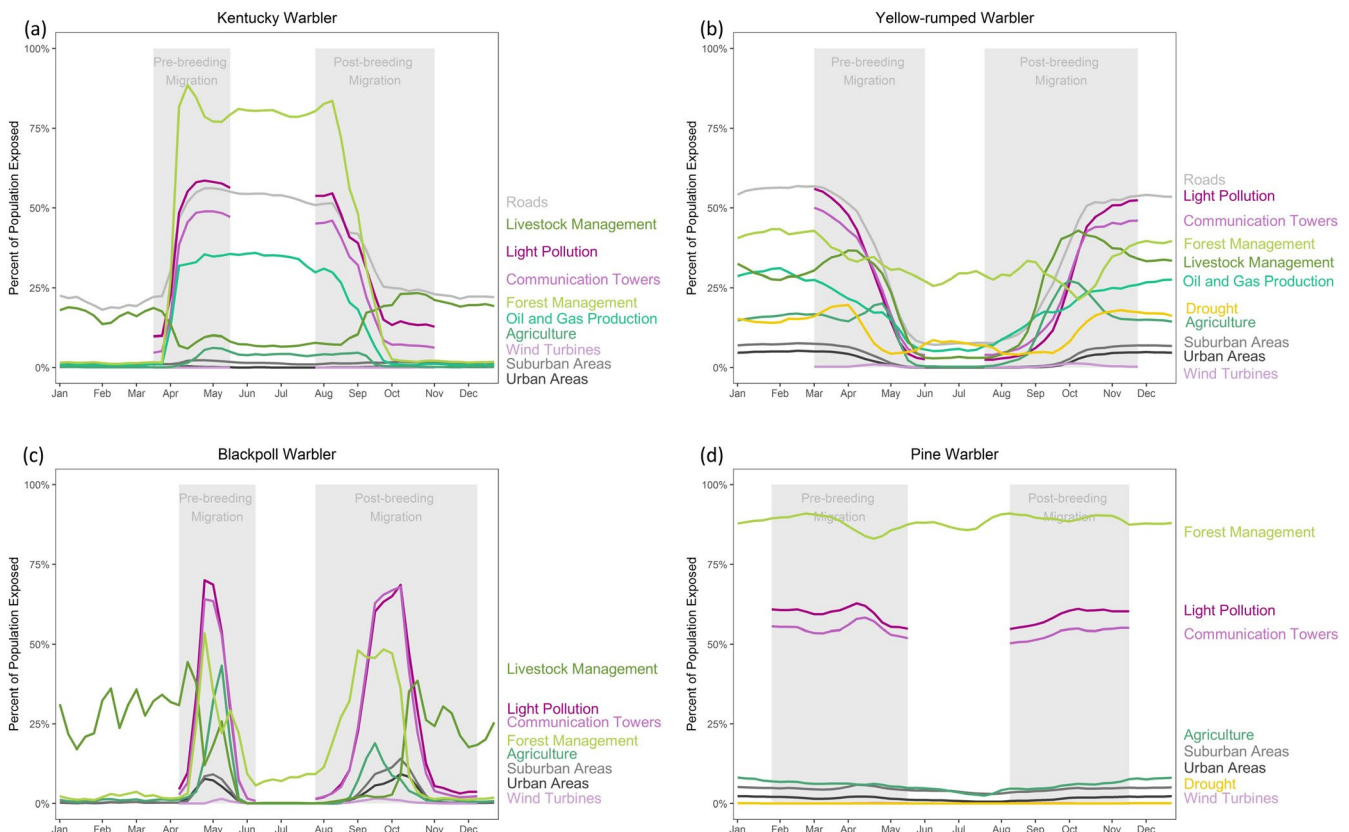


FIGURE 2 | Species' weekly exposure to conservation challenges plotted across the full annual cycle, highlighting example species that are most exposed (a) in the breeding season, (b) in the stationary non-breeding season, (c) during migration, and (d) throughout the year.

during the breeding season (Figure 2a), while other species, like the Yellow-rumped Warbler (*Setophaga coronata*) experienced the least exposure to conservation challenges during the breeding season (Figure 2b). The remaining species showed different patterns, such as the Blackpoll Warbler (*Setophaga striata*), which was most exposed during migration periods (Figure 2c), and the Pine Warbler (*Setophaga pinus*), which was exposed to conservation challenges relatively consistently across all seasons (Figure 2d).

Summarizing across warblers, total seasonal exposure to conservation challenges varied, with the highest exposure occurring during post-breeding migration (Figure 3). Across conservation challenges, warbler species had the highest median total seasonal exposure to roads during all but the breeding season, for which the highest median exposure was to forest management (Figure 3). In addition to roads, during the non-breeding season, warblers had the highest total seasonal exposure to livestock management and drought, while during migration periods, they had the highest total seasonal exposure to light pollution, forest management, and communication towers. Across seasons, warblers had minimal total seasonal exposure to surface water management and wind turbines (Figure 3).

3.2 | Correlation Structure of Warbler Exposure to Conservation Challenges

For each season, there was a substantial correlation among exposure to conservation challenges across species (Figure S2),

warranting further consideration with separate PCAs for each season. For post-breeding migration, the first two principal components cumulatively explained 54% ($p < 0.01$) of the variation in exposure (Table S5). PC 1 described a gradient that represented increasing exposure to collision-related conservation challenges, including communication towers, light pollution, suburban and urban areas, and forest management at one end and greater exposure to livestock management at the other (Figure S3, Table S6). PC 2 described a gradient with greater exposure to agriculture, livestock management, surface water management, wind turbines, and roads at one end and greater exposure to forest management at the other (Figure S3, Table S6).

For pre-breeding migration, we retained the first two principal components ($p < 0.01$), which cumulatively explained 57% of the variation in exposure (Table S5). PC 1 described a gradient of development, with agriculture, suburban areas, light pollution, and communication towers loading especially heavy, along with moderate values for wind turbines and urban areas (Figure S3, Table S6). PC 2 described a gradient with greater exposure to livestock management and drought at one end and greater exposure to communication towers and light pollution at the other end (Figure S3, Table S6).

For the stationary non-breeding season, we retained the first two PCs ($p < 0.01$), which cumulatively explained 52% of the variation in warbler exposure to conservation challenges (Table S5). PC 1 was highly correlated with exposure to all conservation challenges except livestock management and drought. Higher scores for PC 1 represent greater exposure to agriculture, urban

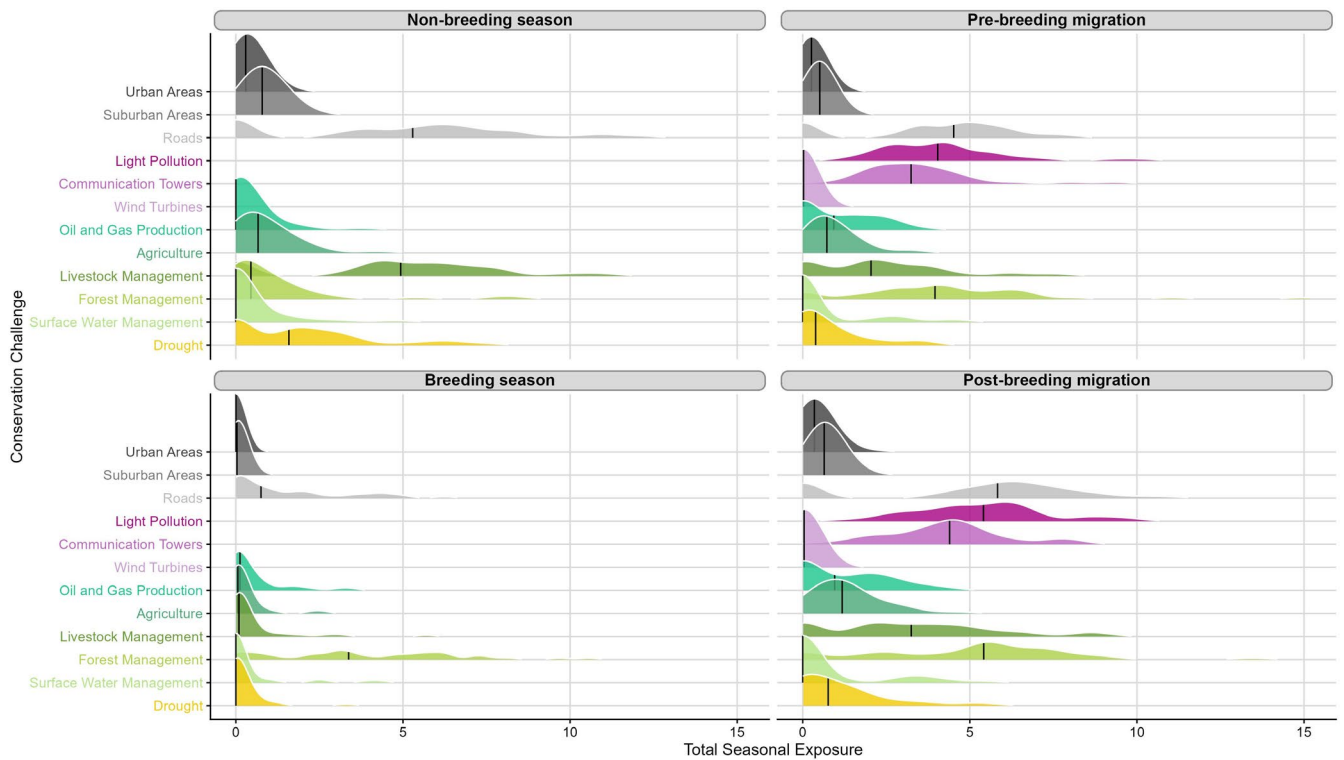


FIGURE 3 | Distribution of the total seasonal exposure (summed across weeks within a season) to each conservation challenge for 43 warbler species throughout the annual cycle: The stationary non-breeding season (top left), pre-breeding migration (top right), breeding season (bottom left), and post-breeding migration (bottom right). Distributions farther to the right represent conservation challenges with greater total seasonal exposure across species. The height of the distributions represents the density of total seasonal exposure values across species, with taller peaks indicating more species with similar exposure (i.e., less interspecies variation). Conservation challenges are grouped by: Urbanized lands (urban areas, suburban areas, roads; gray colors); collision infrastructure (light pollution, communication towers, wind turbines; purple colors); working lands (oil and gas production, agriculture, livestock management, forest management, surface water management; green colors); and drought (yellow). Vertical black lines indicate medians. Collision-related challenges (purple colors) were not relevant to the stationary non-breeding and breeding seasons.

and suburban areas, roads, and forest management (Figure S4, Table S7). PC 2 described an exposure gradient of conservation challenges related to the management of forested and agricultural areas. Low PC 2 scores were characterized by higher exposure to surface water management, drought, and livestock management, and high PC 2 scores had greater exposure to forest management (Figure S4, Table S7).

During the breeding season, the first two principal components explained 61% of the variation in exposure to conservation challenges ($p < 0.01$, Table S5). PC 1 described an increasing gradient of agricultural and development-related challenges. PC 2 described a gradient with greater exposure to livestock management and drought on one end and greater exposure to forest management on the other (Figure S4, Table S7).

3.3 | Seasonal Conservation Challenges and Warbler Population Trends

Across the annual cycle, the variation in warbler population trends was best explained by exposure to conservation challenges during post-breeding migration (42% deviance explained), accounting for more than twice the amount of variation in trends compared to pre-breeding migration (20% deviance explained) or the stationary non-breeding season (20% deviance explained;

TABLE 2 | Statistics from seasonal generalized additive models with warbler species North American Breeding Bird Survey trend estimate as the response variable and retained principal components (PC) as the predictor variables.

Season	PC 1	PC 2	% Total deviance explained
Stationary non-breeding	59.1%	41.0%	19.9%
Pre-breeding migration	15.6%	84.4%	19.5%
Breeding	—	—	< 1%
Post-breeding migration	58.4%	41.6%	41.8%

Note: For PC 1 and PC 2, we report the percent of total deviance explained attributed to each predictor (PC 1 and PC 2) for the interpretation of relative variable contributions to the models (Lai et al. 2024). Because the model for the breeding season explained so little deviance, we did not further evaluate the predictors for that season. See Figures 4–6 for GAM partial regression plots. See Table S5 to reference retained PCs for each season and Tables S6 and S7 and Figures S3 and S4 for the interpretation of each PC, which vary by season. All models were fit with a Gaussian distribution and $k = 10$.

Table 2). Exposure to conservation challenges during the breeding season was not an important predictor of warbler population trends (Table 2). For the post-breeding migration and stationary

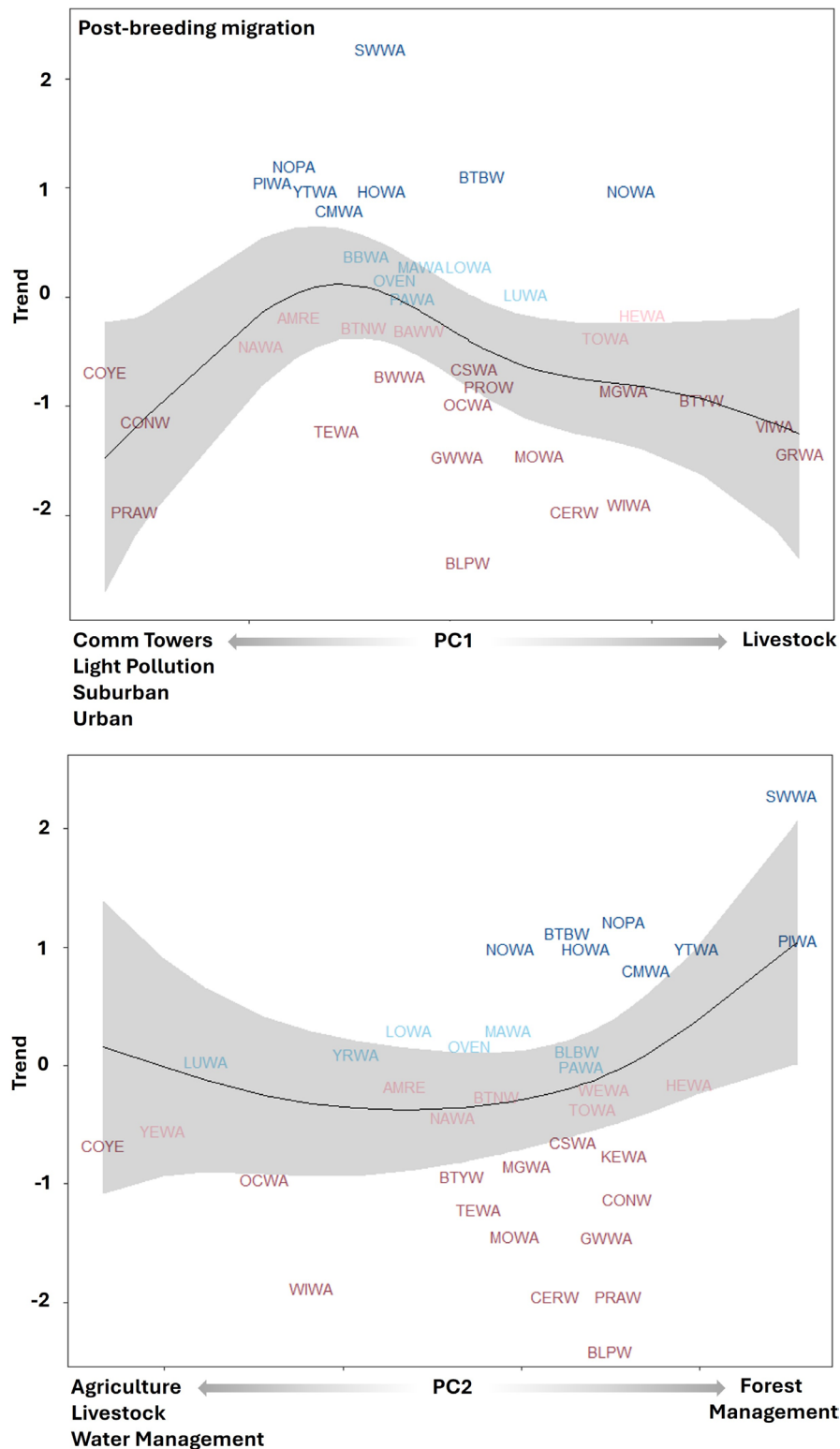


FIGURE 4 | Legend on next page.

non-breeding seasons, PC 1 explained approximately 20% more variation in warbler trends than PC 2. In contrast, PC 2 was substantially more important than PC 1 in the pre-breeding migration model (Table 2).

During post-breeding migration, urbanized, agricultural, and collision landscapes were all associated with decreasing

warbler population trends (Figure 4). Three declining species, Connecticut Warbler (*Oporornis agilis*), Common Yellowthroat (*Geothlypis trichas*), and Prairie Warbler (*Setophaga discolor*), were the most strongly associated with communication towers, light pollution, and urban areas during post-breeding migration. Of the four species most strongly associated with agriculture, livestock, and surface water management in this season, all but

FIGURE 4 | Partial regression plots of generalized additive models (GAM) for post-breeding migration. The full model includes both the retained principal component 1 (PC 1) and principal component 2 (PC 2) as predictor variables. Conservation challenges that are highly correlated (>0.6) with each PC are noted on the x axis. Increasingly dark gray arrows denote the direction of increasing values of the listed conservation challenges. See Figure S3 and Table S6 for additional interpretation of the post-breeding migration principal component analysis of conservation challenges. Warbler population trends estimates were based on the Breeding Bird Survey and are the response variable in the GAM. Trend categories are: Increase $\Rightarrow 0.5$ (dark blue), slight increase $= 0-0.5$ (light blue), slight decrease $= -0.5-0$ (light red), decrease $= < -0.5$ (dark red). Alpha codes for warbler species are as follows, see Table S1 for scientific names: OVEN (Ovenbird), Worm-eating Warbler (WEWA), Louisiana Waterthrush (LOWA), Northern Waterthrush (NOWA), Golden-winged Warbler (GWWA), Blue-winged Warbler (BWWA), Black-and-white Warbler (BAWW), Prothonotary Warbler (PROW), Swainson's Warbler (SWWA), Tennessee Warbler (TEWA), Orange-crowned Warbler (OCWA), Lucy's Warbler (LUWA), Nashville Warbler (NAWA), Virginia's Warbler (VIWA), Connecticut Warbler (CONW), MacGillivray's Warbler (MGWA), Mourning Warbler (MOWA), Kentucky Warbler (KEWA), Common Yellowthroat (COYE), Hooded Warbler (HOWA), American Redstart (AMRE), Cape May Warbler (CMWA), Cerulean Warbler (CERW), Northern Parula (NOPA), Magnolia Warbler (MAWA), Bay-breasted Warbler (BBWA), Blackburnian Warbler (BLBW), Yellow Warbler (YEWA), Chestnut-sided Warbler (CSWA), Blackpoll Warbler (BLPW), Black-throated Blue Warbler (BTBW), Palm Warbler (PAWA), Pine Warbler (PIWA), Yellow-rumped Warbler (YRWA), Yellow-throated Warbler (YTWA), Prairie Warbler (PRAW), Grace's Warbler (GRWA), Black-throated Gray Warbler (BTYW), Townsend's Warbler (TOWA), Hermit Warbler (HEWA), Black-throated Green Warbler (BTNW), Canada Warbler (CAWA), Wilson's Warbler (WIWA).

one had decreasing trends, Lucy's Warbler (*Leiothlypis luciae*). Notably, several species strongly associated with forest management during post-breeding migration, had increasing trends (Figure 4, Figure S3). Conversely, warblers with decreasing trends including Grace's Warbler (*Setophaga graciae*), Virginia's Warbler (*Leiothlypis virginiae*), and Black-throated Gray Warbler (*Setophaga nigrescens*), as well as Wilson's Warbler (*Cardellina pusilla*), had meaningful exposure to livestock management and, to a lesser degree, exposure to drought (Figure 4, Figure S3, Table S6).

Exposure to conservation challenges during the stationary non-breeding season and pre-breeding migration had a relatively even influence on warbler population trends (Table 2). Although the relationship between PC 1 and warbler trends during pre-breeding migration indicates that trends increase with increasing exposure to agriculture, suburban areas, light pollution, communication towers, and wind turbines, the two species with the greatest exposure to these challenges (Common Yellowthroat and Yellow Warbler [*Setophaga petechia*]) were declining (Figure 5). PC 1 was substantially less important for determining trends compared to PC 2 (Table 2), which shows that increasing exposure to livestock management and drought led to decreasing trends (Figure 5). Warbler trends also increased with increasing PC 2 values (Figure 5); however, this was primarily driven by Pine Warbler's exposure to forest management (Figure S3). Three declining species (Yellow Warbler, Common Yellowthroat, and Orange-crowned Warbler [*Leiothlypis celata*]) were associated with agriculture and wind turbines during pre-breeding migration (Figure 5, Figure S3). On the contrary, Northern Parula (*Setophaga americana*) and Pine Warbler, both with increasing trends, were strongly associated with communication towers, light pollution, and forest management (Figure S3). The species most associated with drought during pre-breeding migration, Wilson's Warbler, was in decline (Figure 5). During the stationary non-breeding season, increased exposure to agriculture, urban and suburban areas, and roads (PC 1) led to decreases in some warbler population trends. Several species experiencing steep declines, including Prairie Warbler, Wilson's Warbler, and Common Yellowthroat contributed to this relationship (Figure 6). However, Palm Warbler (*Setophaga palmarum*), Lucy's Warbler, and Yellow-rumped

Warbler were also strongly associated with the same challenges and had slightly increasing trends (Figure 6). Furthermore, several species, including Blackpoll Warbler, Cerulean Warbler (*Setophaga cerulea*), Connecticut Warbler, and several others, had decreasing trends and were associated with low values of these conservation challenges (Figure 6). Increased exposure to forest management (PC 2) was associated with increasing warbler trends, illustrated by Cape May Warbler (*Setophaga tigrina*) and Pine Warbler (Figure 6).

Although some warblers were linked to conservation challenges during the breeding season (Figure S4), we did not find that breeding season exposure to challenges was important for influencing warbler population trends (Table 2). Therefore, we did not interpret the relationships between warbler trends and exposure to conservation challenges during the breeding season.

4 | Discussion

Our results provide hemispheric, full annual cycle spatial information about the conservation challenges associated with population trends of migratory warblers. Specifically, we found that the variation in warbler population trends was best explained by exposure to conservation challenges during post-breeding migration, followed by pre-breeding migration and the stationary non-breeding seasons. Within the post-breeding migratory season, the conservation challenges with the greatest exposure values were roads, light pollution, communication towers, forest management, and livestock management. Because our analysis was framed explicitly on the spatial extent of human activities (as opposed to land cover), the results directly tie to the causes of habitat change. This provides an ability to understand how global changes are impacting broad-scale patterns of biodiversity, as well as serves as decision support for conservation actions. These findings reveal spatiotemporal patterns that can provide guidance for when and where conservation actions may be most effective at mitigating the declines of migratory species. Such information can also be used to identify future research directions, which can further our understanding of the causal nature of human activities on population declines and the impacts

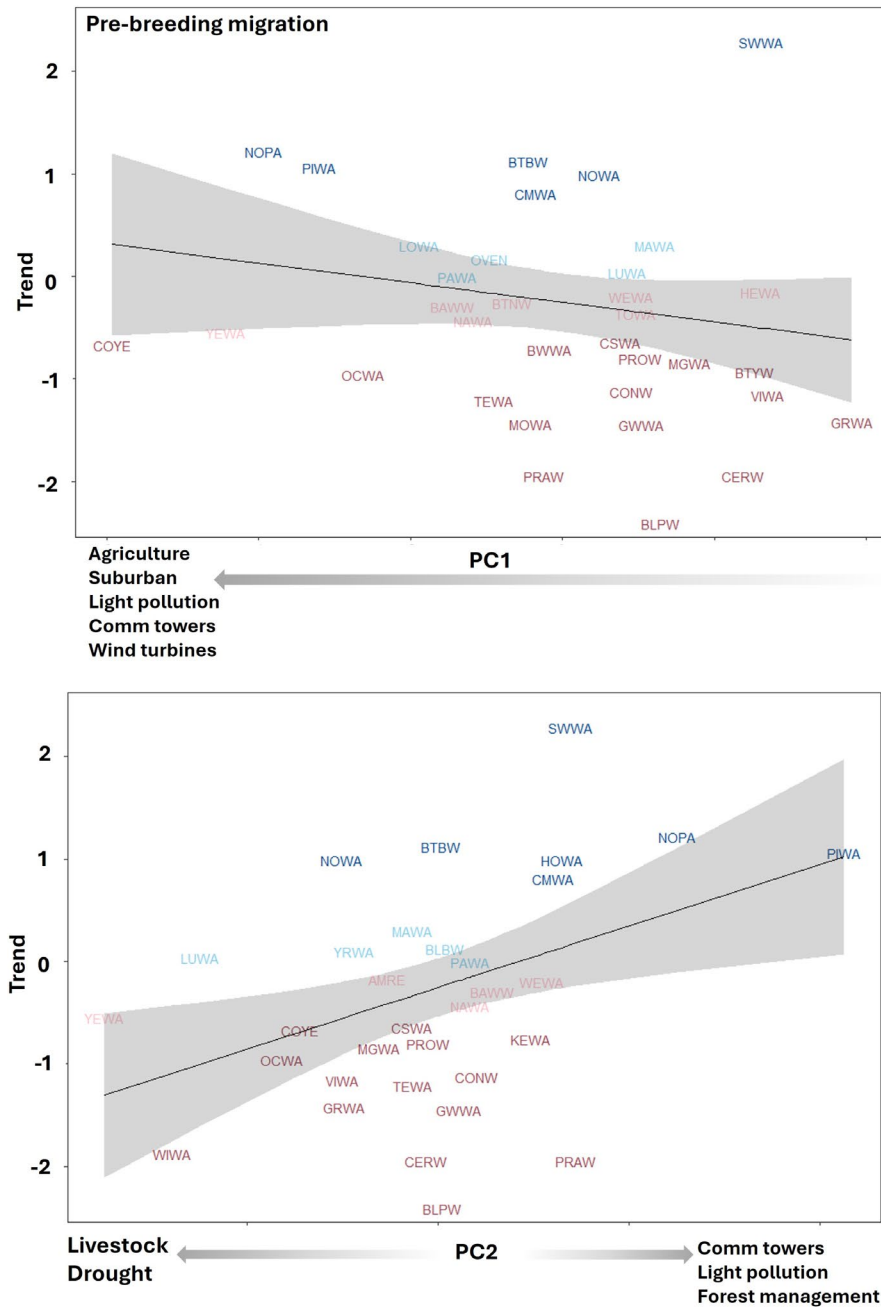


FIGURE 5 | Partial regression plots of generalized additive models (GAM) for pre-breeding migration. The full model includes both the retained principal component 1 (PC 1) and principal component 2 (PC 2) as predictor variables. Conservation challenges that are highly correlated (>0.6) with each PC are noted on the x axis. Increasingly dark gray arrows denote the direction of increasing values of the listed conservation challenges. See Figure S3 and Table S6 for additional interpretation of the pre-breeding migration principal component analysis of conservation challenges. Warbler population trends estimates were based on the Breeding Bird Survey and are the response variable in the GAM. Trend categories are: Increase $\Rightarrow >0.5$ (dark blue), slight increase $= 0-0.5$ (light blue), slight decrease $= -0.5-0$ (light red), decrease $= <-0.5$ (dark red). Alpha codes for warbler species are as follows, see Table S1 for scientific names: OVEN (Ovenbird), Worm-eating Warbler (WEWA), Louisiana Waterthrush (LOWA), Northern Waterthrush (NOWA), Golden-winged Warbler (GWWA), Blue-winged Warbler (BWWA), Black-and-white Warbler (BAWW), Prothonotary Warbler (PROW), Swainson's Warbler (SWWA), Tennessee Warbler (TEWA), Orange-crowned Warbler (OCWA), Lucy's Warbler (LUWA), Nashville Warbler (NAWA), Virginia's Warbler (VIWA), Connecticut Warbler (CONW), MacGillivray's Warbler (MGWA), Mourning Warbler (MOWA), Kentucky Warbler (KEWA), Common Yellowthroat (COYE), Hooded Warbler (HOWA), American Redstart (AMRE), Cape May Warbler (CMWA), Cerulean Warbler (CERW), Northern Parula (NOPA), Magnolia Warbler (MAWA), Bay-breasted Warbler (BBWA), Blackburnian Warbler (BLBW), Yellow Warbler (YEWA), Chestnut-sided Warbler (CSWA), Blackpoll Warbler (BLPW), Black-throated Blue Warbler (BTBW), Palm Warbler (PAWA), Pine Warbler (PIWA), Yellow-rumped Warbler (YRWA), Yellow-throated Warbler (YTWA), Prairie Warbler (PRAW), Grace's Warbler (GRWA), Black-throated Gray Warbler (BTYW), Townsend's Warbler (TOWA), Hermit Warbler (HEWA), Black-throated Green Warbler (BTNW), Canada Warbler (CAWA), Wilson's Warbler (WIWA).

of specific conservation interventions on halting or reversing these declines.

Our results emphasize the importance of the post-breeding migration season (Table 2). Although quantifying mortality during migration has been challenging, there is evidence that birds face high mortality rates during this period (Rockwell et al. 2017; Sillett and Holmes 2002). Our results indicate that the conditions experienced during migration may have population-level impacts across multiple species of migratory birds. During migration, birds are exposed to a greater variety of conservation

challenges as they cover hundreds to thousands of kilometers; post-breeding migration is especially pertinent because of its longer duration (e.g., DeLuca et al. 2019) and influx of inexperienced juvenile migrants (Åkesson et al. 2021).

During the post-breeding migration period, five conservation challenges (roads, light pollution, communication towers, forest management, and livestock management) stood out as having high exposure values across multiple species (Figure 3). The relative importance of these conservation challenges is likely to be highly variable. For example, light pollution and communication

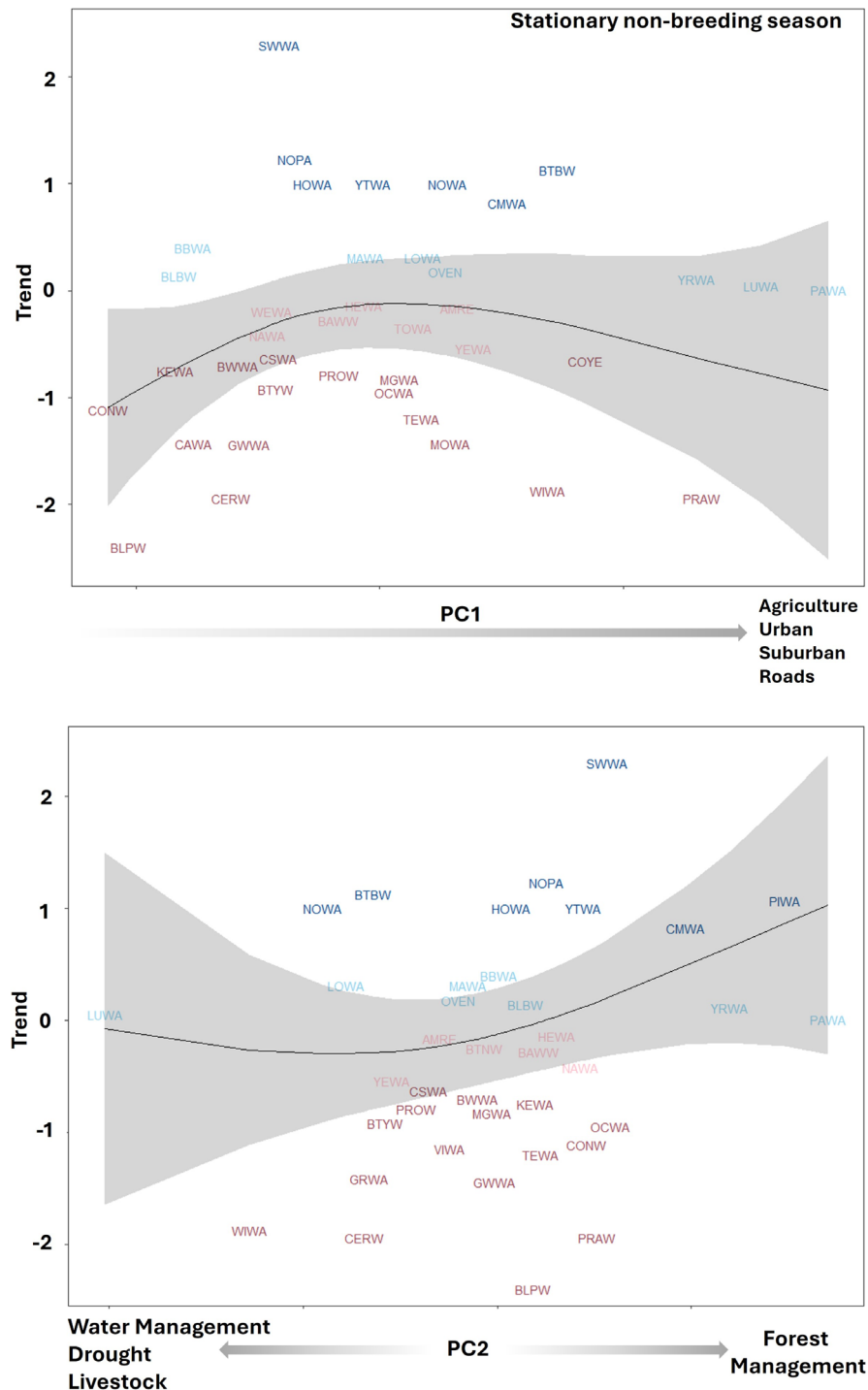


FIGURE 6 | Legend on next page.

FIGURE 6 | Partial regression plots of generalized additive models (GAM) for stationary non-breeding season. The full model includes both the retained principal component 1 (PC 1) and principal component 2 (PC 2) as predictor variables. Conservation challenges that are highly correlated (>0.6) with each PC are noted on the x axis. Increasingly dark gray arrows denote the direction of increasing values of the listed conservation challenges. See Figure S4 and Table S7 for additional interpretation of the stationary non-breeding season principal component analysis of conservation challenges. Warbler population trends estimates were based on the Breeding Bird Survey and are the response variable in the GAM. Trend categories are: Increase $\Rightarrow >0.5$ (dark blue), slight increase $= 0-0.5$ (light blue), slight decrease $= -0.5-0$ (light red), decrease $= <-0.5$ (dark red). Alpha codes for warbler species are as follows, see Table S1 for scientific names: OVEN (Ovenbird), Worm-eating Warbler (WEWA), Louisiana Waterthrush (LOWA), Northern Waterthrush (NOWA), Golden-winged Warbler (GWWA), Blue-winged Warbler (BWWA), Black-and-white Warbler (BAWW), Prothonotary Warbler (PROW), Swainson's Warbler (SWWA), Tennessee Warbler (TEWA), Orange-crowned Warbler (OCWA), Lucy's Warbler (LUWA), Nashville Warbler (NAWA), Virginia's Warbler (VIWA), Connecticut Warbler (CONW), MacGillivray's Warbler (MGWA), Mourning Warbler (MOWA), Kentucky Warbler (KEWA), Common Yellowthroat (COYE), Hooded Warbler (HOWA), American Redstart (AMRE), Cape May Warbler (CMWA), Cerulean Warbler (CERW), Northern Parula (NOPA), Magnolia Warbler (MAWA), Bay-breasted Warbler (BBWA), Blackburnian Warbler (BLBW), Yellow Warbler (YEWA), Chestnut-sided Warbler (CSWA), Blackpoll Warbler (BLPW), Black-throated Blue Warbler (BTBW), Palm Warbler (PAWA), Pine Warbler (PIWA), Yellow-rumped Warbler (YRWA), Yellow-throated Warbler (YTWA), Prairie Warbler (PRAW), Grace's Warbler (GRWA), Black-throated Gray Warbler (BTYW), Townsend's Warbler (TOWA), Hermit Warbler (HEWA), Black-throated Green Warbler (BTNW), Canada Warbler (CAWA), Wilson's Warbler (WIWA).

towers are major causes of collision-related mortality (Van Doren et al. 2021), and it is unlikely there would be a positive outcome for species that are exposed. In contrast, there have been extensive efforts (especially in North America) to develop forest and livestock management practices that benefit birds (Hanberry and Thompson III 2019; Michel et al. 2020), such that the impacts of these conservation challenges are likely to be positive for some species and in some seasons. Interpreting the influence of individual conservation challenges is further complicated because exposure to multiple conservation challenges (e.g., light pollution and communication towers) was correlated. These complexities illustrate the need to better understand the relationship (e.g., the possibility of threshold effects) between the exposure to a conservation challenge and its long-term impact on a species population.

Our results provided little evidence that conservation challenges during the breeding season were associated with warbler population trends. We propose two explanations for this result. First, many of the species in this study have breeding distributions that include vast areas of the boreal forest. While the boreal has certainly been impacted by conservation challenges, such as forest management, roads, and oil and gas development (Matsuoka et al. 2019; Wells et al. 2020), our results suggest that warblers may have thus far been resilient to these changes compared to the challenges they are exposed to during migration and the stationary non-breeding season. However, in the future, these species may become more vulnerable during the breeding season due to anticipated changes throughout the boreal region, especially those that develop as a consequence of rapid warming at northern latitudes (Cadieux et al. 2020). Second, although there is evidence for breeding season factors driving population trends of specific warbler species, such as Kirtland's Warbler (*Setophaga kirtlandii*) (Bocetti et al. 2012) and Connecticut Warbler (Hallworth et al. 2021), there is also evidence for a more general pattern of warbler population trends being tied to variation in adult survival rather than reproductive success (Saracco et al. 2008). Thus, based on current evidence, the impacts of conservation challenges on survival during migration and the stationary non-breeding season should be a focus for conservation actions.

This species-specific hemispheric assessment takes us one step closer to understanding limiting factors. An ongoing challenge

for managers is the identification of meaningful conservation actions to address species declines (Saunders et al. 2021). For example, deforestation, often cited as a major driver of species declines (Rushing et al. 2016), is itself the outcome of a human activity, which can be the result of several different threats, such as logging, agricultural development, or pasture development (Salafsky et al. 2008). By focusing our analysis on conservation challenges, our results translate to root causes of habitat degradation, making them more direct and actionable. For example, Common Yellowthroat stands out for being strongly exposed to agriculture during all four seasons and collision infrastructure and urbanized lands during both migration seasons. Our results align with previous findings for this species demonstrating negative impacts of agricultural practices on abundance and breeding success (Murray and Best 2014; Quinn et al. 2017) and mortality from collision infrastructure (Choi et al. 2020; Loss, Will, Loss, et al. 2014). Looking closely at the variables most associated with trends for each species may guide what conservation challenges researchers and practitioners should consider for a particular species. Further, this examination can shed light on commonalities that can lead to more efficient solutions via conservation aimed at multiple species.

While this approach provides unique information about full annual cycle exposure patterns, it has limitations. First, our approach was restricted to a subset of conservation challenges that can be mapped hemispherically. Some conservation challenges, such as the distribution and abundance of invasive species, are mapped across smaller spatial extents and, thus, cannot be included in hemispheric analyses. While historical data for certain climate variables, such as increasingly severe heat waves, were available hemispherically, the ephemeral nature of these events made them difficult to integrate into our analysis. Ecosystem encroachment, severe and extreme weather events, fire and fire suppression, and problematic native plants and animals were also among the most-cited threats to warblers that we were unable to include in this analysis. Second, our analysis was intended to examine broad-scale ecological signals across multiple species. While this information is invaluable for developing rapid and efficient responses to commonalities in conservation challenges for groups of species, a need remains to invest at the species level to develop detailed and comprehensive

conservation action plans, especially for species with unique ecologies (Rushing et al. 2017). As a result, applying our results to the design of local conservation actions must be done in collaboration with community partners and considering on-the-ground conditions (DeLuca et al. 2023).

Our analysis did not incorporate migratory connectivity, given that we treated each season independently. This limitation may have influenced our results in several ways. First, there may be cumulative impacts of conservation challenges across the annual cycle that are not captured by our analysis (e.g., Buchan et al. 2023). Such patterns may be critical to identify for those species that likely face high levels of exposure on both the breeding and stationary non-breeding grounds, such as high-latitude, long-distance breeders (Kramer et al. 2018; Saunders et al. 2025), which experience rapidly warming temperatures in one season and compounding challenges during extended migrations because of carryover effects. Second, migratory connectivity may contribute to understanding spatial variation in exposure to conservation challenges—and thus population trends—across the full range of a species (Kramer et al. 2023). Finally, we acknowledge that there is variation in the uncertainty of species trend estimates (Stanton et al. 2019) and we did not account for this uncertainty. Future work should incorporate migratory connectivity into spatially explicit threat assessments and trend uncertainty to further refine estimates of population trend drivers.

This work takes us one step closer to estimating species risk across global scales. Our analysis effectively integrates two components of risk—exposure (the presence of a species or system that could be adversely affected) and vulnerability (the predisposition to be adversely affected based on sensitivity and lack of adaptive capacity). The third element of risk, hazard (the potential occurrence of a damaging physical event), is included as well, although our work did not estimate the likelihood of such events, which would be necessary to directly estimate population impacts from conservation challenge exposure. Linking this approach to estimates of migratory connectivity (Knight et al. 2021; Saunders et al. 2025; Webster et al. 2002) would further inform which populations are most at risk. As the bird conservation community moves toward full annual cycle conservation planning (Marra et al. 2015), our results suggest the importance of considering a comprehensive set of conservation challenges at broad spatial extents when examining potential causes and challenging assumptions about population declines across species. A relevant next step would be the mapping of “conservation challenge hotspots” for warblers. Such an analysis could combine exposure values across conservation challenges for a single species, filtered to those challenges that are associated with a negative trend, and weighted by local abundance. Across declining species, the resulting layers could then be weighted by population trend and further integrated to create a spatially explicit identification of places where the most individuals experience the greatest exposure to challenges associated with declining trends.

Our results demonstrate that spatial patterns of exposure to conservation challenges, especially during migration, are correlated with population trends of migratory warblers. This work builds upon a growing understanding of the role that human activities

across the globe play in contributing to migratory bird population trends (Buchan et al. 2022). Public policies can be an efficient and effective mechanism for reducing the impacts of human activities across large geographies and contributing to the reversal of population declines. One example is the North American Wetlands Conservation Act (NAWCA) of 1989 in the United States that authorizes funding for the protection, enhancement, restoration, and management of wetland ecosystems to provide habitat for waterfowl. At least 50% of funding under NAWCA is allocated to wetland conservation in Canada and Mexico to address the loss of wetlands across the continent due to broad-scale human activities and ensure waterfowl have safe habitat across the full annual cycle. The legislation, supported by large federal appropriations and other funding sources, has contributed to reversing declining trends of waterfowl populations between 1930 and 1990. Over the last 50 years, waterfowl were one of the few groups that showed population increases (Rosenberg et al. 2019) illustrating the role of broad policies for driving change at scale.

The hemispheric scale of conservation challenges underscores the need for multinational efforts to address them and reverse population declines of migratory birds (Baldwin 2011; UNEP-WCMC 2024). The Americas Flyways Framework developed under the CMS enumerates several strategic goals and actions needed to conserve migratory birds and their habitats in the Western Hemisphere. Priority actions include active cooperation and coordination among partners and initiatives to reduce direct pressures and threats to migratory birds and advance actions that provide benefits for migratory birds. Our findings provide valuable insight into the conservation challenges migratory birds are exposed to across the hemisphere, and in which seasons their exposure is greatest, to inform and advance these initiatives.

Author Contributions

Nathaniel E. Seavy: conceptualization, data curation, investigation, methodology, project administration, supervision, validation, writing – original draft, writing – review and editing. **Melanie A. Smith:** conceptualization, data curation, investigation, methodology, project administration, supervision, validation, visualization, writing – original draft, writing – review and editing. **William V. DeLuca:** data curation, formal analysis, investigation, methodology, software, validation, visualization, writing – original draft, writing – review and editing. **Erika J. Knight:** data curation, formal analysis, methodology, software, validation, visualization, writing – original draft, writing – review and editing. **Chad J. Witko:** data curation, investigation, validation, writing – review and editing. **Sarah P. Saunders:** methodology, validation, writing – original draft, writing – review and editing. **Lotem Taylor:** data curation, formal analysis, methodology, software, writing – review and editing. **Daniela Linero-Triana:** data curation, validation, writing – review and editing. **Jorge Velásquez-Tibatá:** data curation, validation, writing – review and editing. **Chad B. Wilsey:** conceptualization, funding acquisition, supervision, writing – review and editing. **Jill L. Deppe:** conceptualization, funding acquisition, supervision, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.14939292> as well as on GitHub at https://github.com/audubongit/Warbler_conservation_challenges.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.